Ex.No.-1.1-Design and Draft the given 2D Sketches in modelling software.

AIM:

To sketch the given drawing with dimensions using fusion 360 tool.

REQUIREMENTS:

1.Autodesk fusion 360

2.Windows 11

3.16 GB of RAM (integrated graphics recommend 2 GB or more)

4.2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:

1. Open a new file and then select the sketch icon.
2. Select the plane for sketching.
3. Use sketching features (line, circle, arc, rectangle, polygon, etc.,) to draw the given sketch.

Center Circles & Central Features:

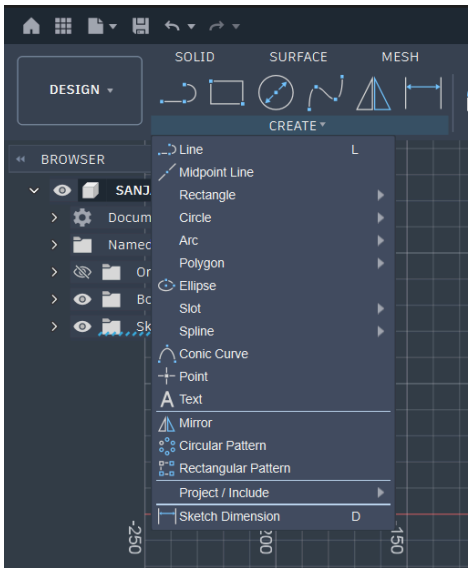
- **Central Star / Multi-lobe Profile:** Inner lobes have a radius of **R6**.
- **Outer Diameter Arc:** Diametral reference of **Ø40** and **Ø66** around the central body.

Side Lobe / Circular Projections (Left & Right):

- **Center-to-Center Distance (Horizontal): 47 mm** on both the left side and right side from the central vertical axis (total span = 94 mm).
- **Vertical Offsets from Centerline:**
    - **26 mm** upward vertical distance to the upper reference.
    - **24 mm** downward vertical distance to the lower feature.
- **Left/Right Outer Loops:**
    - Outer circular arcs of radius **R7**.
    - Inner holes with a diameter of **Ø10**.
    - Connecting neck fillets/radii of **R10**.

Bottom & Transitional Curves:

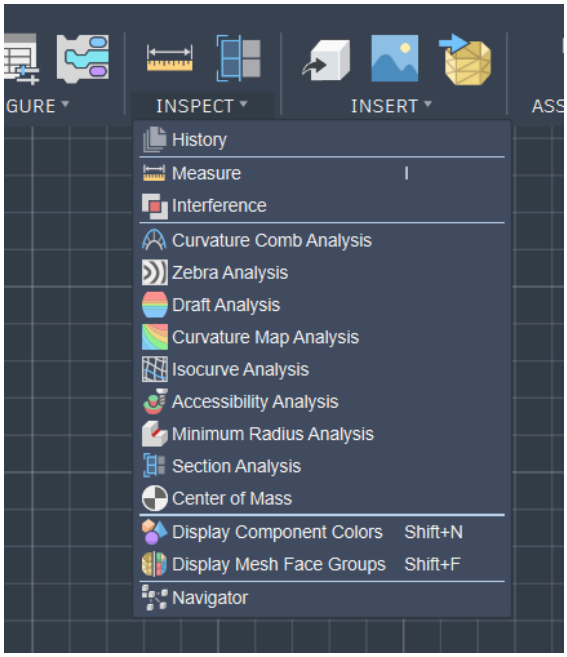
- Bottom central hole diameter of **Ø10**.
- Surrounding lower hub/flange diameter of **Ø20**.
- Transitional connecting curves/fillets with radii of **R20**.
- Upper-left curved profile transition of radius **R20**.



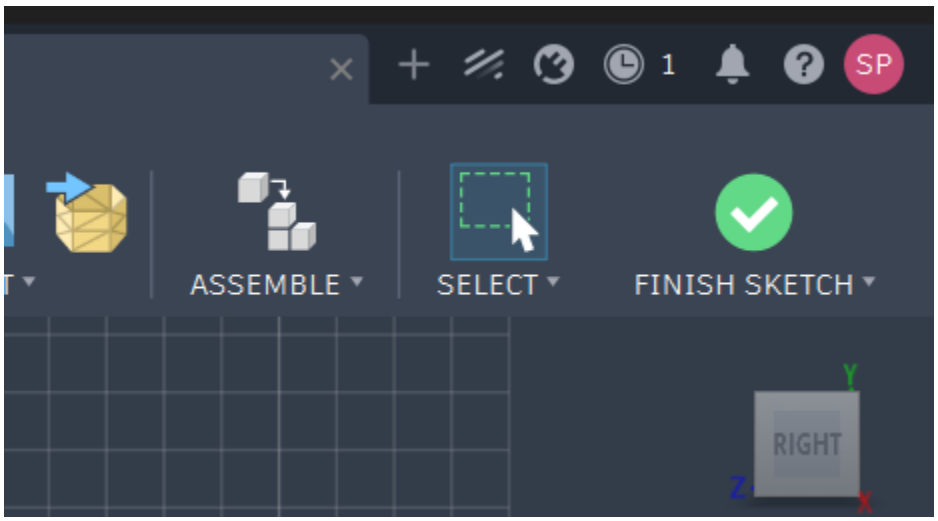
4. Use sketch modifying features (cut, trim, mirror, move, etc.,) to edit the sketch.



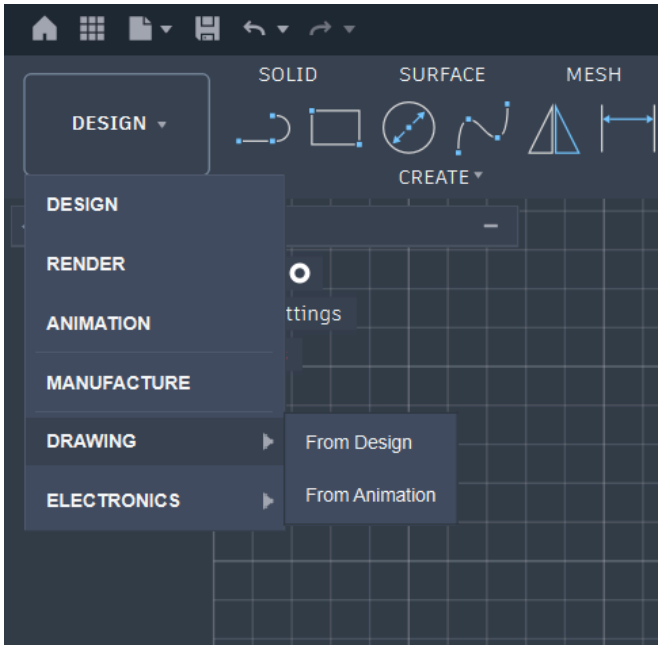
5. Use inspect feature to measure the dimensions of the sketch.

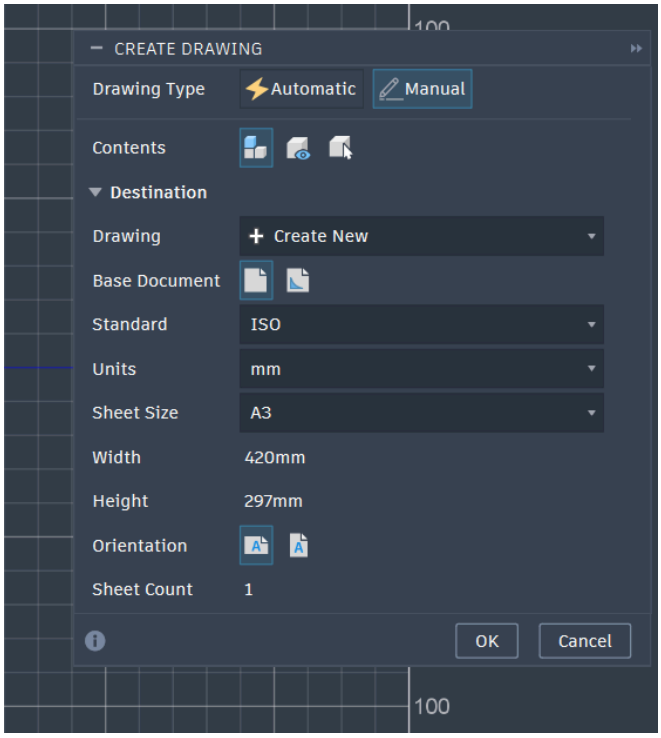


6. After finishing the sketch use the finish sketch option to finalize the sketch.

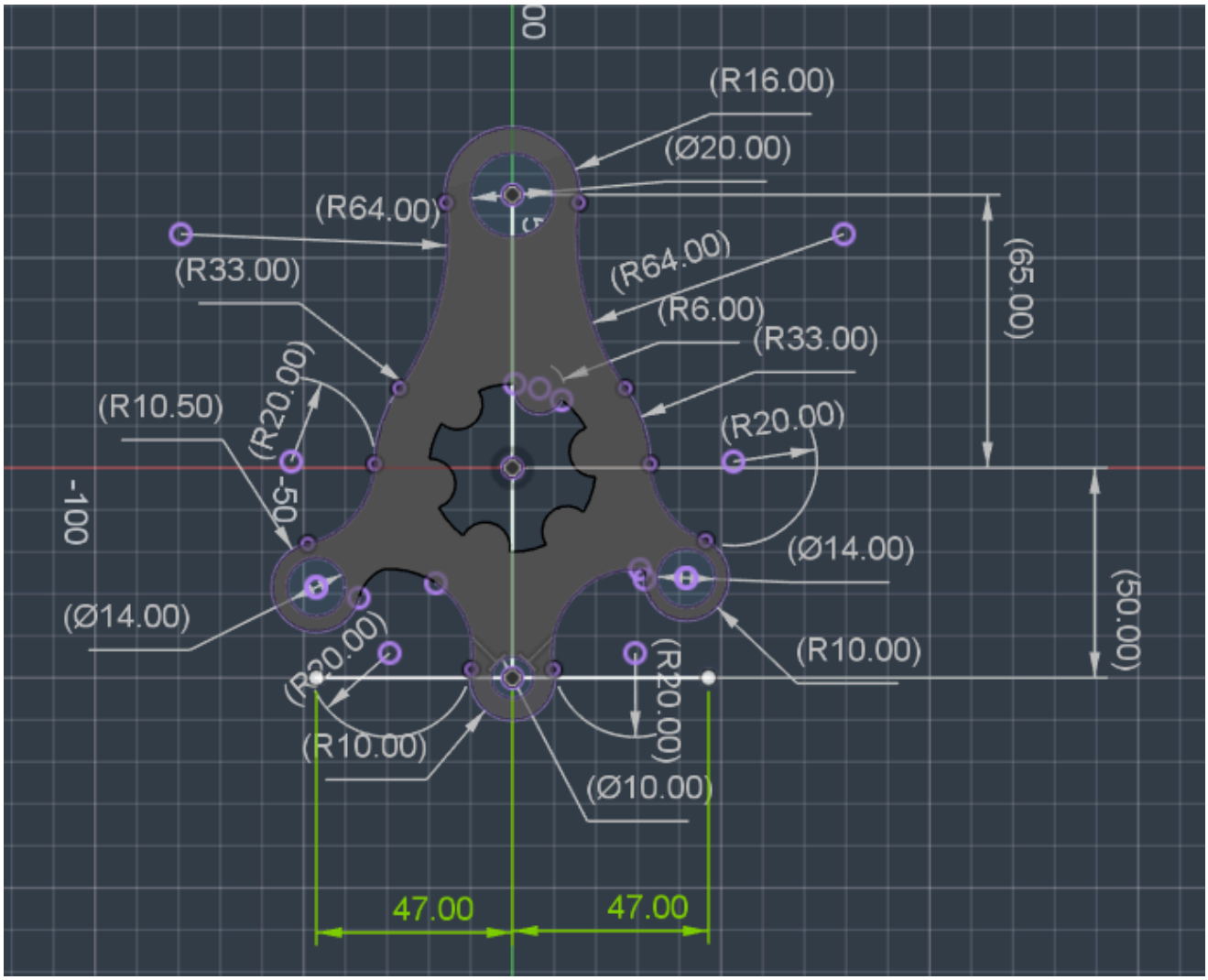


7. After finishing the sketch, switch to drawing mode to plot the sketch in ISO standard.

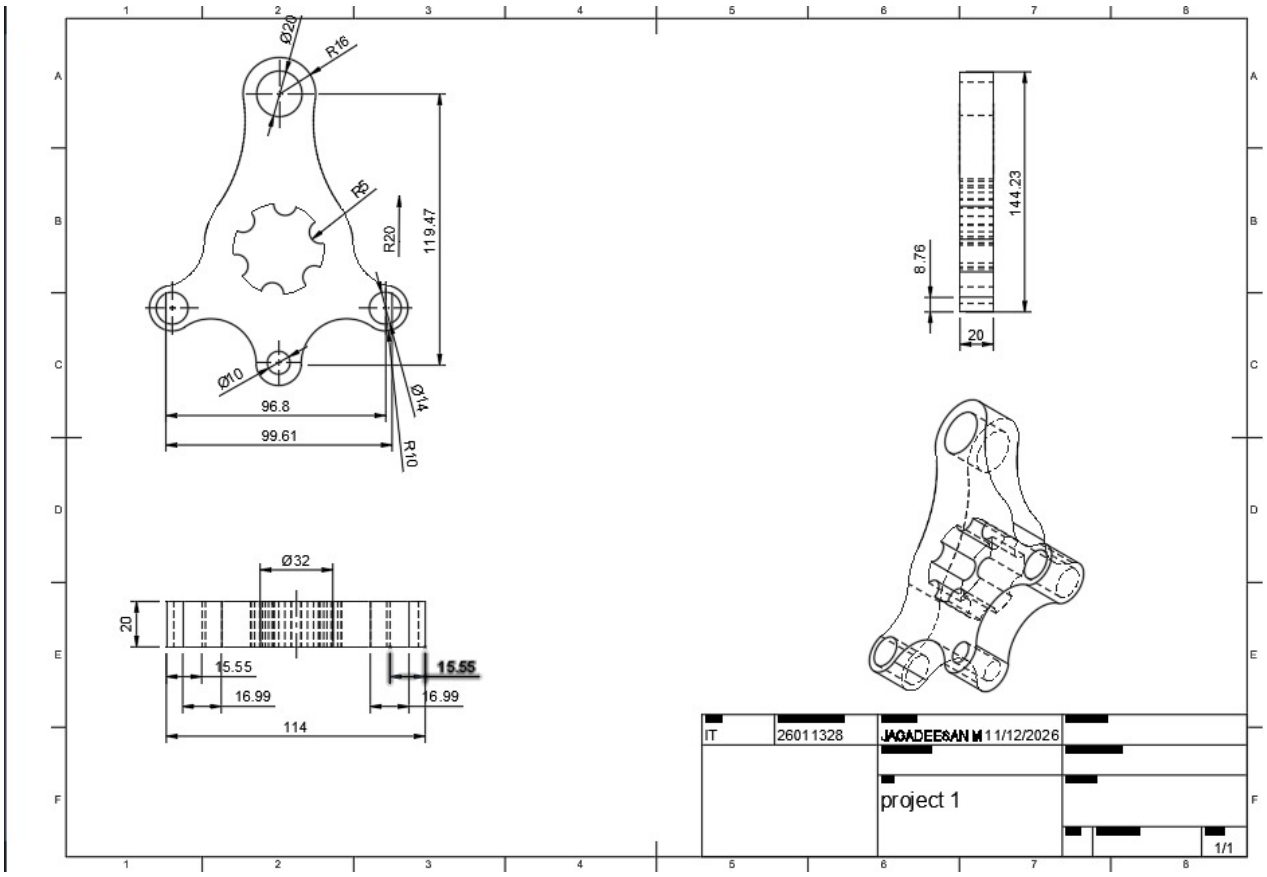




SKETCHS:


(R16.00)
(Ø20.00)
(R64.00)
(R33.00)
(R64.00)
(R6.00)
(R33.00)
(65.00)
(R20.00)
(R10.50)
(R20.00)
-50
-100
(Ø14.00)
(50.00)
(Ø14.00)
(R10.00)
(R20.00)
(R20.00)
(R10.00)
(Ø10.00)
47.00
47.00

OUTPUT:



RESULT:

Thus the given sketch is drawn and drafted using the fusion 360 tool.